Smart Home Health Monitor

PPP Presentation

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Problem Statement

Most indoor air quality systems are inaccurate and incomplete.

- Sensors in air purifiers measure already cleaned air
- Many systems only monitor PM2.5
- Other important environmental factors are ignored

Why This Problem Is Important

Poor indoor air quality can cause:

- Sleep problems
- Reduced concentration and productivity
- Long-term respiratory issues

People spend most of their time indoors, especially while sleeping.

Proposed Solution

Build an intelligent IoT air quality monitoring system that:

- Measures air at breathing height
- Monitors multiple environmental parameters
- Automatically controls air purifiers
- Provides health and environmental insights

Project Domain / Focus

This project focuses on:

- Sensor integration
- Home automation
- IoT communication
- Actuator control (smart plug)
- Software dashboard and mobile interface

System Overview

Sensors → ESP32 → Cloud / MQTT → Dashboard → Smart Plug → Air Purifier

Hardware Components

- ESP32 Microcontroller
- MH-Z19B CO_2 Sensor
- PMS5003 PM2.5 Sensor
- SGP40 VOC Sensor
- AHT20 Temperature & Humidity Sensor
- OLED Display
- Smart Plug (Actuator)

Software Architecture

- Embedded firmware (Arduino / PlatformIO)
- Web/Mobile dashboard.
- Firebase or cloud database
- Python for machine learning

Learning With AI

Software Topic

Mobile/Web Application Development

AI will help with:

- UI design
- Debugging
- API integration
- Real-time dashboard development

Learning With AI

Hardware Topic

Multi-Sensor Integration & Calibration

AI will help with:

- Datasheet interpretation
- Communication debugging
- Calibration techniques
- Firmware development

Time Usage – Iteration 1

Learning & Foundation

- Understand sensor communication
- Make individual sensors work
- ESP32 firmware setup
- Basic data collection and display

Time Usage – Iteration 2

Feature Development

- Smart purifier automation
- Web dashboard
- Data logging system
- Alert notifications
- Sleep quality data correlation

Project Timeline

- Week 4 – MVP and PPP Demo
- Week 8 – Data Collection Phase Complete
- Week 12 – Machine Learning Model
- Week 14 – Final Presentation

Final Goals

- Working smart air quality prototype
- Real-time dashboard and mobile interface
- Automated purifier control
- Environmental data analysis
- Portfolio project
- Research poster or publication potential

Risk Management

- Sensor accuracy → Calibration and validation
- Hardware delays → Parallel software development
- Internet reliability → Local automation fallback
- Time constraints → Prioritize core features

Current Progress

✓ Project planning completed

✓ GitHub repositories created

✓ Requirements and architecture defined

Next Step:

- Order hardware
- Begin individual sensor testing

Expected Impact

This project combines IoT, automation, and data analysis to improve indoor air quality awareness and health.

Questions?